

Raw LIAR Files
(train.tsv, valid.tsv, test.tsv)



Load & Concatenate
(tag rows with split; unify schema)



Structural Validation
(dtype check, whitespace/empty-string cleanup)



Missing Value Analysis
(per-column count & percentage)



Duplicate Detection
(check on id and statement text)



Null Handling
Drop core-null rows + Impute categorical "Unknown"



Feature Engineering
Text stats, credit-history aggregates, binary target



Correlation Analysis
(numeric feature relationships)



Feature Scaling
Min-Max normalization + Standardization



Final Engineered Dataset
(41 columns, 12,789 rows, 0 missing values)